

NGUYEN BA LONG

Thu Duc, Ho Chi Minh City, Viet Nam

+84 396 978 406 longnguyen132005@gmail.com [LinkedIn](#) [GitHub](#) [Portfolio](#)

Summary

Final-year computer science student at UIT with a solid foundation in deep learning, computer vision, and multimodal AI. Experienced in designing custom neural architectures, building agentic RAG workflows (LangGraph), and engineering scalable hybrid retrieval systems. Seeking an AI Engineer/Research Intern position to contribute research-driven solutions to frontier AI challenges.

Education

University of Information Technology (UIT) - VNUHCM

Sep. 2023 - Jun. 2027 (Expected)

Bachelor of Science in Computer Science

Ho Chi Minh City, Vietnam

- **GPA:** 8.27/10.0
- **Relevant Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Mining, Big Data.

Technical Skills

Languages: Python, C/C++, SQL, JavaScript

AI/ML & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers

Generative AI & RAG: LangChain, LangGraph, Prompt Engineering, Vector Search, Hybrid Retrieval, Reranking

Computer Vision & Multimodal: OpenCV, PaddleOCR, Image Segmentation, VLM, Dataset Annotation

Data Processing: ETL Pipelines, Data Cleaning, Feature Engineering, Distributed Processing

Backend & Databases: FastAPI, Flask, RESTful APIs, PostgreSQL, Vector & Search Databases, Docker, Git

Projects

Temporal RAG for UIT Course Video Q&A | *LangGraph, Qwen2-VL, Whisper, FastAPI, Chroma*

[GitHub](#)

- Built an agentic RAG pipeline with LangGraph to answer questions across 55 lecture videos, integrating query guardrails and HyDE transformation to improve search context.
- Extracted slide text using Qwen2-VL and audio transcripts using Whisper-large-v3, synchronizing multimodal content into structured chunks with exact video timestamps.
- Implemented a hybrid retrieval engine combining BM25 keyword search and Chroma vector search (multilingual-e5-large) via Reciprocal Rank Fusion (RRF, $k = 60$) and BGE-Reranker.
- Developed an asynchronous FastAPI backend to return answers with clickable video timestamps; benchmarked the system on 90 test queries via RAGAS, reaching 0.96 Faithfulness and 0.89 Answer Relevancy.

AIC Multimodal Video Retrieval System | *PaddleOCR, SigLIP2, YOLOv11, Milvus, Elasticsearch, Redis*

[GitHub](#)

- Engineered a multimodal ingestion pipeline for 726 videos across 5 modalities (SigLIP2, Whisper, PaddleOCR, Florence-2, YOLOv11); handled dataset preparation and frame annotation.
- Built an OCR & ASR extraction pipeline using PaddleOCR (PP-OCRv5) and Whisper-large-v3, applying OpenCV image preprocessing and Vietnamese prompt engineering to extract on-screen text and speech transcripts.
- Developed a hybrid multi-stream retrieval engine combining Milvus dense vector search (SigLIP2) and Elasticsearch BM25 sparse search via Reciprocal Rank Fusion (RRF, $k = 60$), fine-tuned with Cross-Encoder (BGE-Reranker).
- Integrated query decomposition (Gemini API / rule-based fallback), interactive Rocchio Relevance Feedback, and Redis caching to achieve sub-second search latency across KIS, Visual Q&A, and TRAKE queries.

HTS-MultiResUNet: Dual-Attention Hybrid CNN-Transformer | *TensorFlow, Keras, Scikit-learn, OpenCV* [GitHub](#)

- Engineered a hybrid CNN-Transformer architecture for multimodal biomedical image segmentation (Fluorescence, Electron, Dermoscopy, Endoscopy), coupling Squeeze-and-Excitation (SE) channel attention with a Multi-Head Self-Attention (MHSA) bottleneck to capture long-range contextual dependencies.
- Constructed a robust training pipeline using 5-fold cross-validation (Scikit-learn), strict image normalization, and gradient stabilization techniques to handle severe spatial class imbalances across 4 distinct datasets; evaluated model performance using IoU and Dice metrics.
- Formulated an Edge-Aware Focal Tversky Loss (EA-FTL) with a discrete Sobel operator to preserve fine anatomical boundaries, outperforming the baseline MultiResUNet by 5.83% in Jaccard Index (IoU) and achieving superior Dice scores.

Languages

Vietnamese: Native

English: Professional Reading Proficiency; TOEIC Speaking & Writing 250/400.